STA 2002

Lecture 15

Nonparametric Confidence interval

Sign test

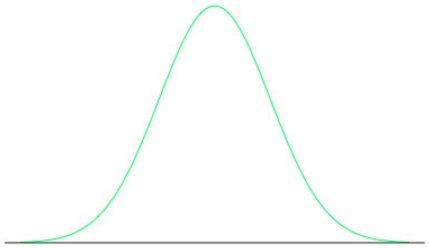
Reminder about CI

Confidence interval for a population parameter θ :

A random interval calculated from the sample, that **contains θ with some specified probability**.

"If $\boldsymbol{\mu}$ takes this value, the likelihood of $\bar{X}$ is that"

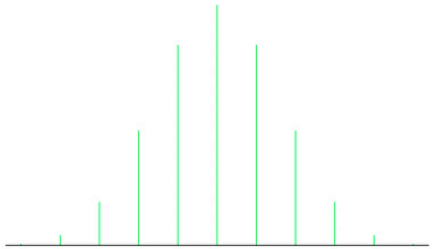
$P(\bar{X}|\mu)$:



Now what we have is:

$P(Data|m)$

m is the median.



Ex:

$$P(-z(\alpha/2) \leq Z \leq z(\alpha/2)) = 1 - \alpha$$

Based on CLT, $(\overline{X} - \mu)/\sigma_{\overline{X}}$ approximately follows a standard normal distribution. So,

$$P\left(-z(\alpha/2) \leq \frac{\overline{X} - \mu}{\sigma_{\overline{X}}} \leq z(\alpha/2)\right) \approx 1 - \alpha$$

Rearrange,

$$P(\overline{X} - z(\alpha/2)\sigma_{\overline{X}} \leq \mu \leq \overline{X} + z(\alpha/2)\sigma_{\overline{X}}) \approx 1 - \alpha$$

Confidence Interval for median

Given the ordered data, we would like to $P(X_{(i)} < m < X_{(j)}) \approx 1 - \alpha$, where $i < j$ and m is the median.

Suppose that we have the sample $X_1, \ldots, X_5$. Then we have the order statistics $X_{(1)}, X_{(2)}, \ldots, X_{(5)}$. Our goal is to estimate the median $m = \pi_{0.5}$. A natural point estimator of m is $X_{(3)}$.

Suppose we use the interval $(X_{(1)}, X_{(5)})$ as a confidence interval for m. We consider the probability

$$\begin{aligned} P(X_{(1)} < m < X_{(5)}) &= 1 - P(X_{(1)} \geq m) - P(X_{(5)} \leq m) \\ &= 1 - P(X_1 \geq m, X_2 \geq m, \ldots, X_5 \geq m) - P(X_1 \leq m, X_2 \leq m, \ldots, X_5 \leq m) \\ &= 1 - \prod_{i=1}^{5} P(X_i \geq m) - \prod_{i=1}^{5} P(X_i \leq m) \\ &= 1 - 0.5^5 - 0.5^5 = \frac{15}{16} = 94\%. \end{aligned}$$

In other words, $(X_{(1)}, X_{(5)})$ is a 94% C.I. for m.

Once we have data, say $x_1 = 8.7, x_2 = 0.1, x_3 = 4.4, x_4 = 3.1, x_5 = 6.0$, the above 94% C.I. is $(0.1, 8.7)$.

In general, if we have a random sample $X_1, \ldots, X_n$ from an unknown continuous distribution with median m, then

$$\begin{aligned}
P(X_{(1)} < m < X_{(n)}) &= 1 - P(X_{(1)} \geq m) - P(X_{(n)} \leq m) \\
&= 1 - P(X_1 \geq m, X_2 \geq m, \ldots, X_n \geq m) - P(X_1 \leq m, X_2 \leq m, \ldots, X_n \leq m) \\
&= 1 - \prod_{i=1}^{n} P(X_i \geq m) - \prod_{i=1}^{n} P(X_i \leq m) \\
&= 1 - \left(\frac{1}{2}\right)^n - \left(\frac{1}{2}\right)^n = 1 - \left(\frac{1}{2}\right)^{n-1}.
\end{aligned}$$

In other words, $(X_{(1)}, X_{(n)})$ is a $1 - \left(\frac{1}{2}\right)^{n-1}$ C.I. for m.

Suppose now the sample size is $n = 16$. What is the confidence level of the random interval $(X_{(4)}, X_{(11)})$ for m?

Let W be the number of X_i that are less than m. In terms of indicator functions, it is

$$W = \sum_{i=1}^{16} \mathbf{1}_{(X_i<m)} \sim Bin(16, 1/2),$$

"indicator function"
1(TRUE)=1
1(FALSE)=0

since each X_i are i.i.d. and $P(X_i < m) = 0.5$.

$$\begin{aligned}
P(X_{(4)} < m < X_{(11)}) &= P(\text{at least 4 and at most 10 of } X_1, \ldots, X_{16} < m) \\
&= P(W = 4, 5, \ldots, 10) \\
&= \sum_{k=4}^{10} \binom{16}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{16-k} \\
&= \sum_{k=4}^{10} \binom{16}{k} \left(\frac{1}{2}\right)^{16} \\
&= 0.8843.
\end{aligned}$$

$(X_{(4)}, X_{(11)})$ is a 88.43% C.I. for m.

In general, suppose we have a random sample $X_1, \ldots, X_n$ from an unknown continuous distribution with median m.

Let W be the number of X_i that are less than m. In terms of indicator functions, it is

$$W = \sum_{i=1}^{n} \mathbf{1}_{(X_i<m)} \sim Bin(n, 1/2).$$

Then for $i < j$,

$$\begin{aligned} P(X_{(i)} < m < X_{(j)}) &= P(\text{at least i and at most j-1 of } X_1, \ldots, X_n < m) \\ &= P(W = i, i+1, \ldots, j-1) \\ &= \sum_{k=i}^{j-1} \binom{n}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{n-k} \\ &= \sum_{k=i}^{j-1} \binom{n}{k} \left(\frac{1}{2}\right)^n \end{aligned}$$

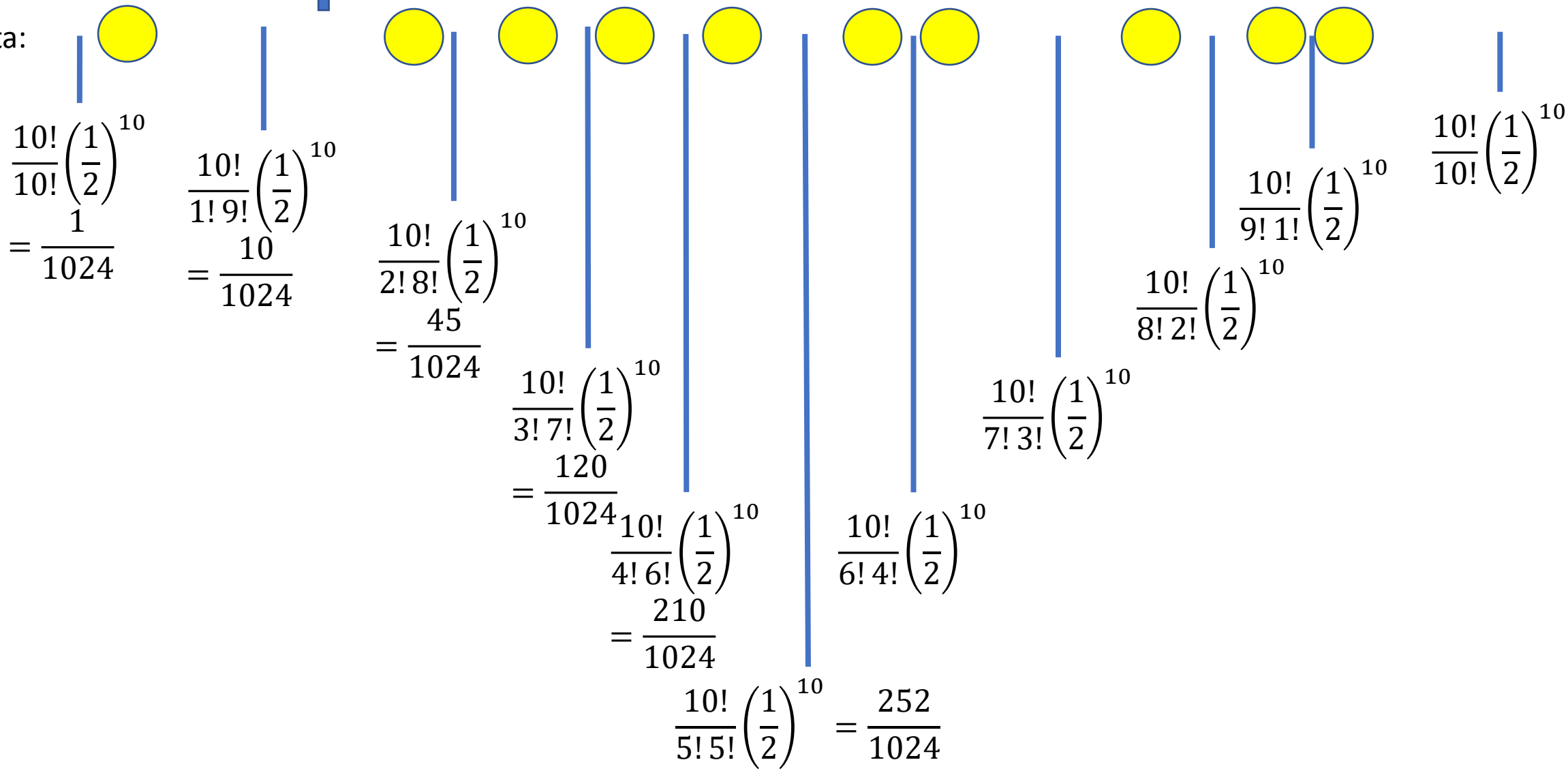
In other words, $(X_{(i)}, X_{(j)})$ is a $\sum_{k=i}^{j-1} \binom{n}{k} \left(\frac{1}{2}\right)^n$ C.I. for m.

CI for the median

What's the data likelihood if the median is in this interval? $(i = 1, j = 2)$

$$\frac{10!}{1!9!}\left(\frac{1}{2}\right)^1\left(\frac{1}{2}\right)^9 = \frac{10!}{1!9!}\left(\frac{1}{2}\right)^{10} = \frac{10}{1024}$$

Data:



CI for the median

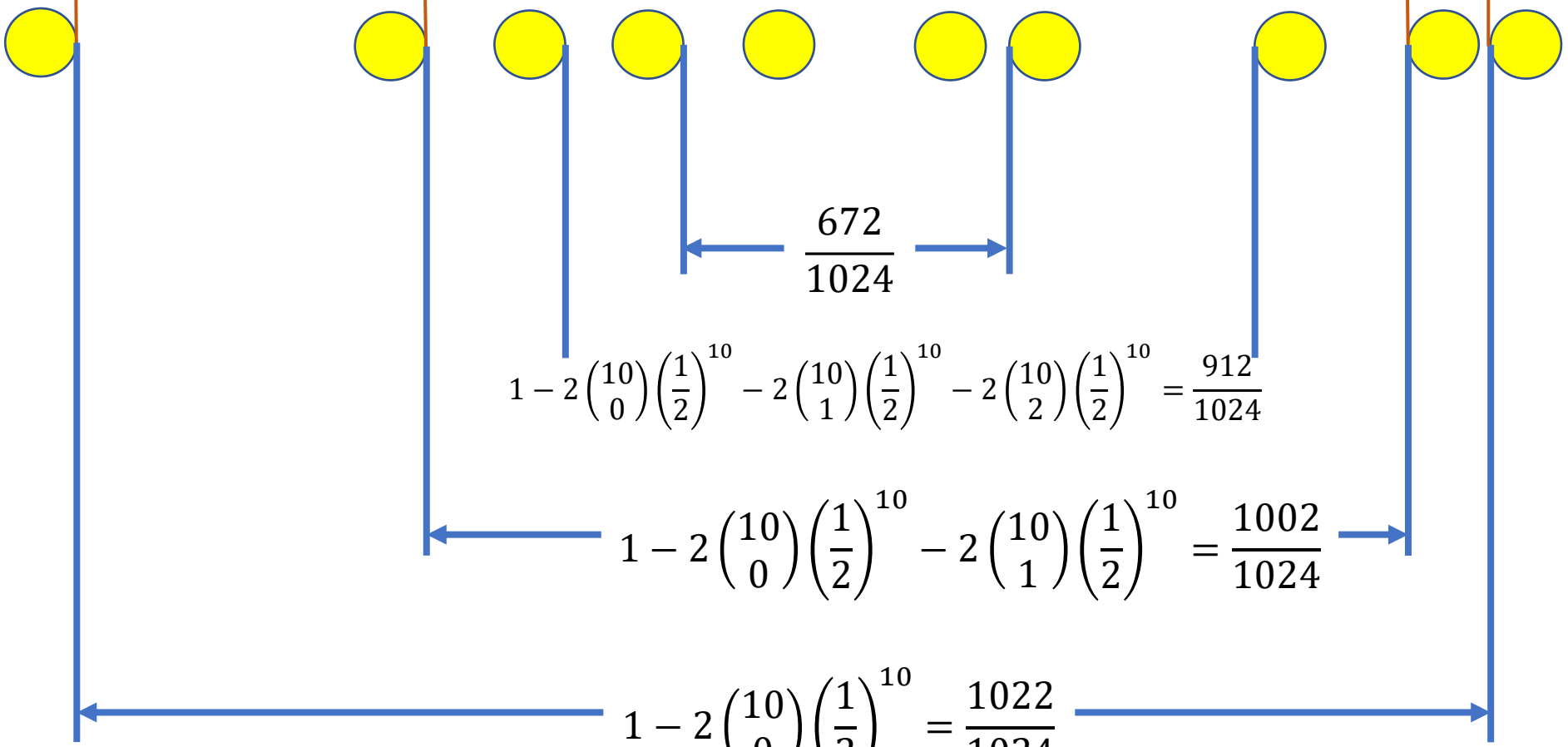
Toss a fair coin 10 times. At least 1 tail; at least 1 head

$$Prob\left(X_{(1)} < m < X_{(10)}\right) = \sum_{k=1}^{9} \binom{10}{k} \left(\frac{1}{2}\right)^{k} \left(\frac{1}{2}\right)^{10-k} = 1 - 2\left(\frac{1}{2}\right)^{10}$$

Toss a fair coin 10 times. At least 2 tails; at least 2 heads

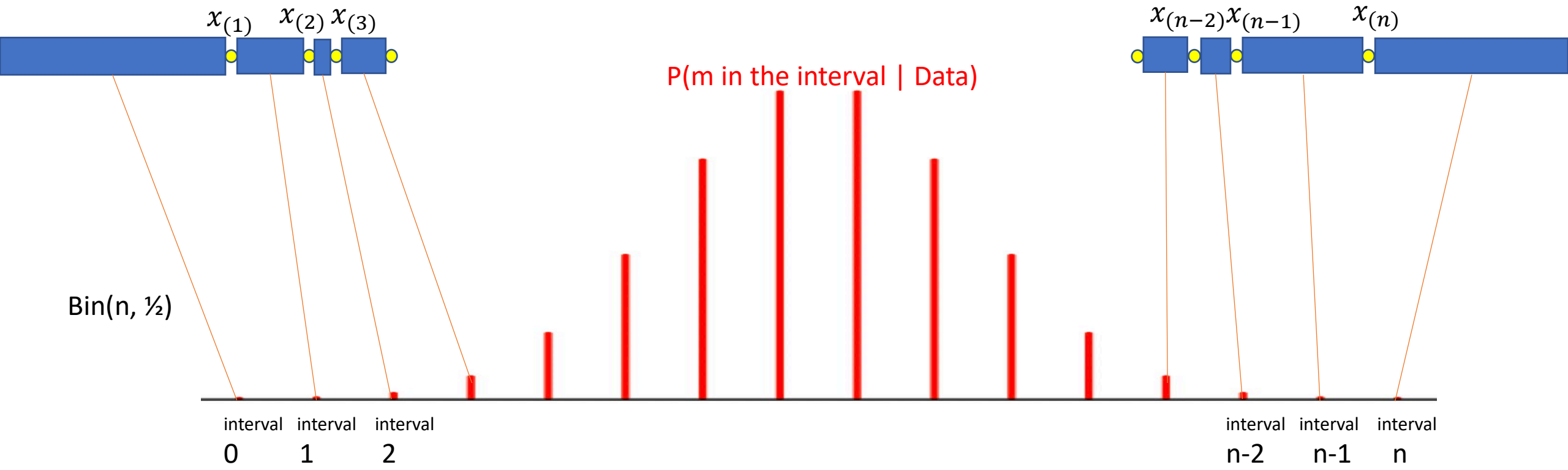
$$Prob\left(X_{(2)} < m < X_{(9)}\right) = \sum_{k=2}^{8} \binom{10}{k} \left(\frac{1}{2}\right)^{k} \left(\frac{1}{2}\right)^{10-k}$$

Data:



CI for the median

So the calculation for the data likelihood for the intervals boils down to binomial probabilities.



$$Interval\ (x_{(1)}, x_{(n)}):\ \ Confidence\ level \sum_{k=1}^{n-1} \binom{n}{k} \left(\frac{1}{2}\right)^n$$

$$Interval\ (x_{(2)}, x_{(n-1)}):\ \ Confidence\ level \sum_{k=2}^{n-2} \binom{n}{k} \left(\frac{1}{2}\right)^n$$

$$Interval\ (x_{(3)}, x_{(n-2)}):\ \ Confidence\ level \sum_{k=3}^{n-3} \binom{n}{k} \left(\frac{1}{2}\right)^n$$

.....

CI for the median

0.40 1.80 2.08 2.42 3.11 3.14 3.37 3.61 3.91 4.59 5.74 6.58 10.91 11.74 15.83 19.76

(0.4, 19.76) confidence: 0.99997

(1.8, 15.83) confidence: 0.99948

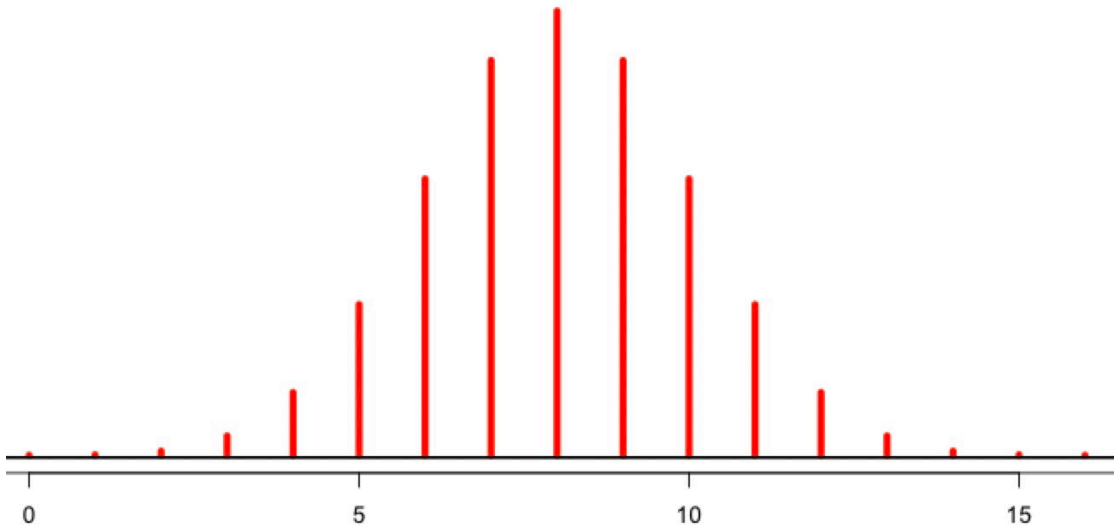
(2.08, 11.74) confidence: 0.99582

(2.42, 10.91) confidence: 0.97873

(3.11, 6.58) confidence: 0.92319

(3.14, 5.74) confidence: 0.78989

In R, $P(X_{(i)} < m < X_{(n-i+1)}) = \sum_{k=i}^{n-i} \binom{n}{k} \left(\frac{1}{2}\right)^n$
$= pbinom(n-i, n, .5)$- $pbinom(i-1, n, .5)$




```
 0 1.5e-05
 1 2.4e-04
 2 1.8e-03
 3 8.5e-03
 4 2.8e-02
 5 6.7e-02
 6 1.2e-01
 7 1.7e-01
 8 2.0e-01
 9 1.7e-01
10 1.2e-01
11 6.7e-02
12 2.8e-02
13 8.5e-03
14 1.8e-03
15 2.4e-04
16 1.5e-05
```

CI for the median

How about normal approximation?

$$W = \sum_{i=1}^{n} 1(x_i - m < 0) \sim binomial\left(n, \frac{1}{2}\right)$$

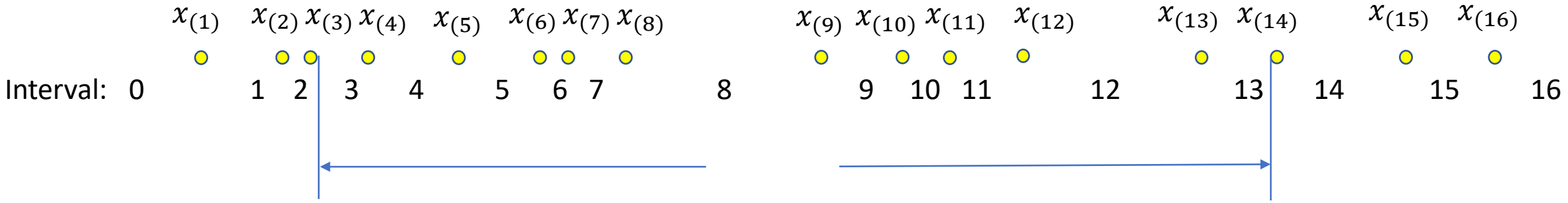
$$W \sim N\left(\frac{n}{2}, \frac{1}{2} \times \frac{1}{2} \times n\right)$$

$$\frac{W - \frac{n}{2}}{\sqrt{n/4}} \sim N(0,1)$$

0.40 1.80 2.08 2.42 3.11 3.14 3.37 3.61 3.91 4.59 5.74 6.58 10.91 11.74 15.83 19.76

For interval (2.08, 11.74), $3 \le W \le 13$
Use **continuity correction** for the ranks 3 & 13 (see next slide)

$$Prob(2.5 < W < 13.5)$$
$$= Prob\left(\frac{2.5 - 8}{2} < z < \frac{13.5 - 8}{2}\right)$$
$$= 1 - 2\Phi(-2.75)$$
$$= 0.994$$



Continuity correction is used when you use a continuous probability distribution to approximate a discrete probability distribution, e.g., when you want to use the normal to approximate a binomial.

- If $P(X=n)$ use $P(n - 0.5 < X < n + 0.5)$
- If $P(X > n)$ use $P(X > n + 0.5)$
- If $P(X \leq n)$ use $P(X < n + 0.5)$
- If $P(X < n)$ use $P(X < n - 0.5)$
- If $P(X \geq n)$ use $P(X > n - 0.5)$

Discrete	Continuous
$x = 6$	$5.5 < x < 6.5$
$x > 6$	$x > 6.5$
$x \leq 6$	$x < 6.5$
$x < 6$	$x < 5.5$
$x \geq 6$	$x > 5.5$

CI for the median

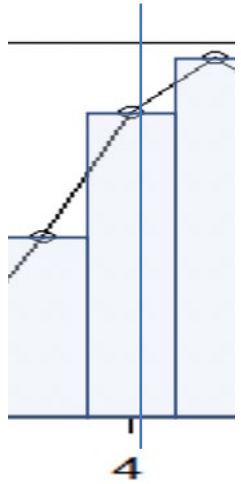
For 100(1-α)% CI, find integer *W* such that its interval contains the largest *W* satisfying $\dfrac{W-\frac{n}{2}}{\sqrt{n/4}} \le -z_{\alpha/2}$

0.40 1.80 2.08 2.42 3.11 3.14 3.37 3.61 3.91 4.59 5.74 6.58 10.91 11.74 15.83 19.76

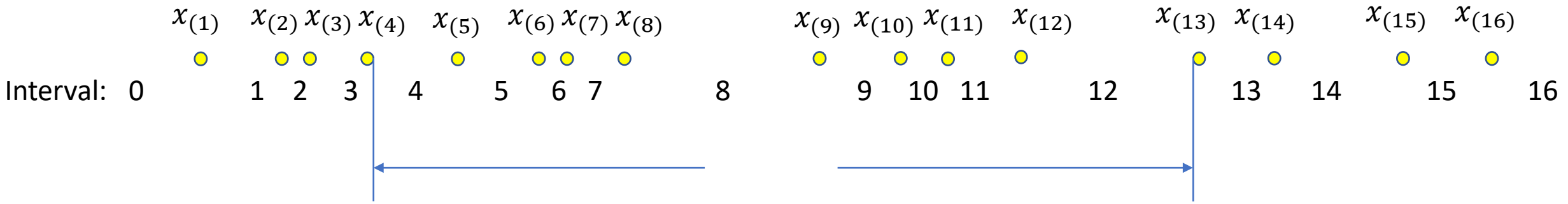
For 95% confidence,

$$\frac{W-\frac{16}{2}}{\sqrt{16/4}} \le -z_{\alpha/2} = -1.96$$

$$W \le 8 - 1.96 \times 2 = 4.08$$



The interval is (2.42, 10.91). Its actual coverage is 0.978, larger than 0.95.



CI for other quantiles

Goal: obtain a confidence interval for the *p*th quantile π_p, where $p \in [0,1]$, e.g., $\pi_{0.25}$.

Again the calculation for the likelihood for each interval boils down to binomial probabilities.

Let W be the number of X_i that are less than $\pi_{0.25}$. In terms of indicator functions, it is

$$W = \sum_{i=1}^{16} \mathbf{1}_{(X_i < \pi_{0.25})} \sim Bin(16, 0.25),$$

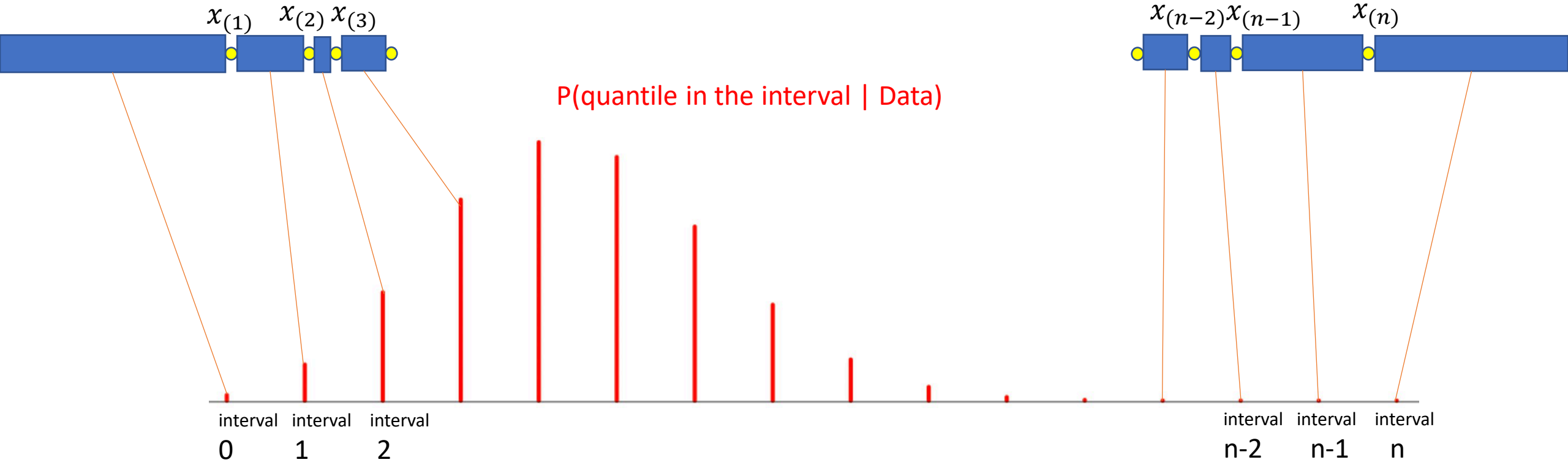
since each X_i are i.i.d. and $P(X_i < \pi_{0.25}) = 0.25$. Consider the probability

$$\begin{aligned} P(X_{(3)} < \pi_{0.25} < X_{(10)}) &= P(\text{at least 3 and at most 9 of } X_1, \ldots, X_{16} < \pi_{0.25}) \\ &= P(W = 3, 4, \ldots, 9) \\ &= \sum_{k=3}^{9} \binom{16}{k} (0.25)^k (0.75)^{16-k}. \end{aligned}$$

CI for other quantiles

Goal: obtain a confidence interval for the *p*th quantile π_p, where $p \in [0,1]$.

Again the calculation for the likelihood for each interval boils down to binomial probabilities.



$$Interval\ (x_{(i)}, x_{(j)}):\ \ Confidence\ level \sum_{k=i}^{j-1} \binom{n}{k} (p)^k\ (1-p)^{n-k}$$

CI for other quantiles

1.33 1.74 2.07 2.76 2.88 2.96 3.58 3.82 4.11 4.22 4.38 5.04 5.06 7.20 7.81 8.47

For $\pi_{0.3}$, note that (16+1)x0.3=5.1. We consider

$Interval\ (x_{(1)}, x_{(10)})$: *(1.33, 4.22)*

$Confidence\ level$

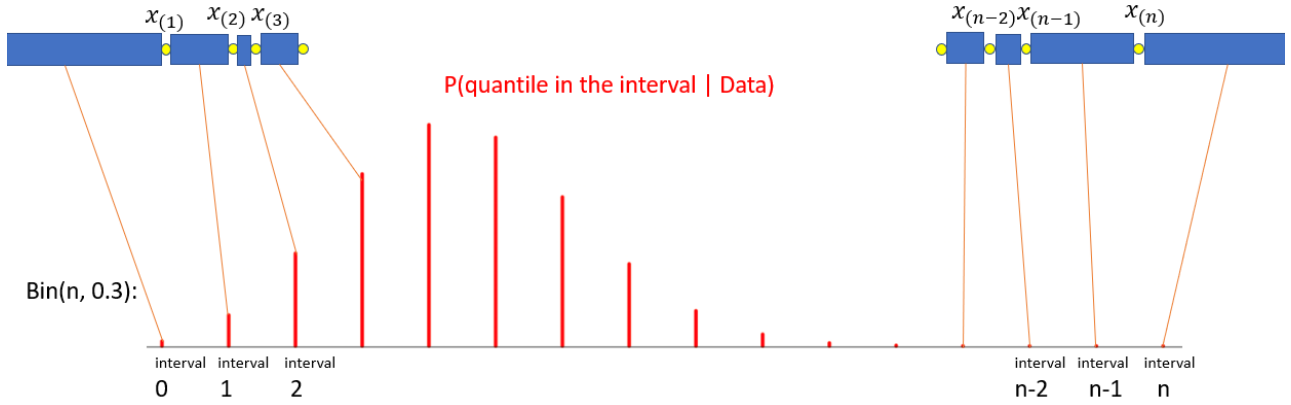
$\sum_{k=1}^{9} \binom{16}{k} (0.3)^k\ (1-0.3)^k = 0.9895$

```
 0 3.32e-03
 1 2.28e-02
 2 7.32e-02
 3 1.46e-01
 4 2.04e-01
 5 2.10e-01
 6 1.65e-01
 7 1.01e-01
 8 4.87e-02
 9 1.85e-02
10 5.56e-03
11 1.30e-03
12 2.32e-04
13 3.06e-05
14 2.81e-06
15 1.61e-07
16 4.30e-09
```

Normal approximation:

$W \sim N(16 \times 0.3, 0.3 \times 0.7 \times 16)$
$W \sim N(4.8, 3.36)$

$Prob(0.5 < W < 9.5)$
$= Prob\left(\frac{0.5-4.8}{\sqrt{3.36}} < z < \frac{9.5-4.8}{\sqrt{3.36}}\right)$
$= Prob(-2.34 < z < 2.564)$
$= 0.985$



Sign test

One-sample **distribution-free** test on median

$$H_0: m = m_0$$

$$H_1: m \neq m_0$$

Each sample has ½ chance to fall above the median.

Under the null, the number of samples that are $< m_0$ follows a binomial distribution

$$\sum_{i=1}^{n} 1(x_i - m_0 < 0) \sim binomial(n, \frac{1}{2})$$

"indicator function"
1(TRUE)=1
1(FALSE)=0

Count samples below m_0. Then this becomes a one-sample test of binomial probability.

Sign test

Data:

0.44	−0.06	0.43	−0.16	−2.13
1.15	1.08	5.67	−4.97	0.11

Hypotheses:

$H_0: m=0$
$H_1: m\neq 0$



Equivalent to:

$n=10$
$x=4$

$H_0: p=0.5$
$H_1: p\neq 0.5$

Sign test

Normal approximation:

$$z = \frac{0.4 - 0.5}{\sqrt{0.5(1-0.5)/10}} = -0.63$$

$$p\ value = 2 \times Prob(Z < -0.63) = 0.52$$

(this is without continuity correction)

With correction:

$$z = \frac{0.45 - 0.5}{\sqrt{0.5(1-0.5)/10}} = -0.316$$

$$p\ value = 2 \times Prob(Z \leq -0.316) = 0.75$$

Without approximation (binomial test):

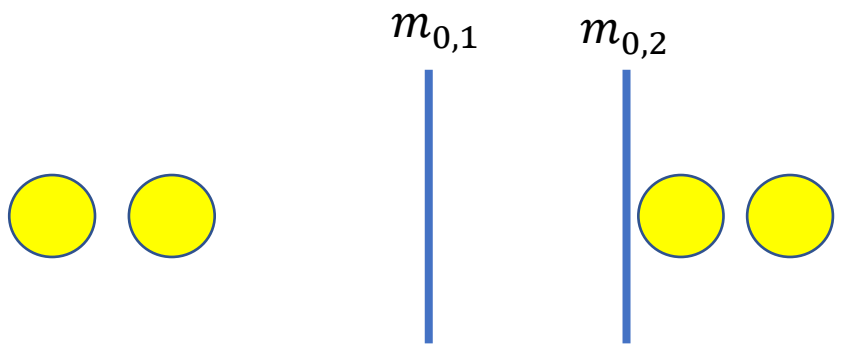
$$p\ value = 2 \times P(X \leq 4) = 0.7539$$

```
> binom.test(4,10)

        Exact binomial test

data:  4 and 10
number of successes = 4, number of trials = 10, p-value = 0.7539
alternative hypothesis: true probability of success is not equal to 0.5
95 percent confidence interval:
 0.1215523 0.7376219
sample estimates:
probability of success
                   0.4
```

Wilcoxon signed rank test



The sign tests for $H_0: m = m_{0,1}$ and $H_0: m = m_{0,2}$ yield the same p-value. But is this reasonable?

One major objection to the sign test is that it does not take into account the magnitude of the differences $x_1 - m_0, \dots, x_n - m_0$.

The Wilcoxon signed rank test aims to address this problem (we skip the details).

In R, we use the function wilcox.test to conduct the test.

The test can also be used to nonparametrically compare two distributions, known as the Mann-Whitney test.